

# Vol 6 Chapter 8 — Phase Transitions

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## Chapter 8 — Phase Transitions

### Level 1 — one sentence

Phase transitions — water boiling, iron magnetizing, helium becoming superfluid — are substrate K-type reorganizations at critical points where the system's macroscopic state changes qualitatively (order parameter switches from zero to nonzero, or vice versa), and Landau's symmetry-breaking framework describes the universal phenomenology while substrate-mechanism identifies which K-type cluster restructuring is happening at each transition.

### Level 2 — graduate-physicist precision

#### 8.1 What is a phase transition

A phase transition is a singular change in the thermodynamic state of a system as a control parameter (typically temperature, pressure, or magnetic field) crosses a critical value.

Two main types: - First-order: discontinuous change in first derivatives of free energy (entropy, volume). Has latent heat. Examples: solid-liquid, liquid-gas at non-critical points. - Second-order (continuous): continuous order parameter that vanishes at  $T_c$ ; divergent susceptibilities. No latent heat. Examples: ferromagnet at Curie temp, superfluid transition, liquid-gas at critical point.

The Ehrenfest classification (1933) labels transitions by which derivative becomes discontinuous; modern usage emphasizes the order parameter structure.

#### 8.2 Order parameters

An order parameter  $\phi$  is a quantity that: - Vanishes in the high-symmetry “disordered” phase - Becomes nonzero in the low-symmetry “ordered” phase - Approaches zero continuously (second-order) or jumps (first-order) at the transition

Examples: - Ferromagnet: magnetization  $\vec{M}$  - Liquid-gas: density difference  $\rho_l - \rho_g$  - Superfluid: complex condensate wavefunction  $\psi$  - Superconductor: BCS gap  $\Delta$  - Crystallization: structure factor amplitude

Substrate-mechanism reading: order parameter is the substrate's K-type cluster amplitude. Above  $T_c$ : substrate K-types fluctuate independently; cluster amplitude averages to zero. Below  $T_c$ : substrate K-types correlate into clusters with macroscopic amplitude.

#### 8.3 The Ising model

The Ising model is the canonical lattice model of phase transitions. Spins  $s_i = \pm 1$  on a lattice, with Hamiltonian:

$$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i$$

$J > 0$  favors aligned neighbors (ferromagnetic);  $h$  is external field.

Order parameter: magnetization  $m = (1/N) \sum_i \langle s_i \rangle$ .

In 1D: no phase transition at finite  $T$  (Ising 1925 — thermal fluctuations destroy long-range order).

In 2D: phase transition at  $k_B T_c/J = 2/\ln(1 + \sqrt{2}) \approx 2.269$  (Onsager 1944 — exact solution, one of the most celebrated results in stat mech).

In 3D: phase transition at  $k_B T_c/J \approx 4.512$  (numerical; no exact solution).

Substrate reading: the Ising spins are substrate Pin(2) K-types coupled to nearest-neighbor substrate K-types via  $J$ . Below  $T_c$ , the substrate's K-type couplings dominate over thermal fluctuations; spins align macroscopically.

#### 8.4 Mean-field theory (Curie-Weiss)

For the Ising model: replace neighbor interactions with their mean. Each spin sees an effective field  $h_{\text{eff}} = h + Jzm$  where  $z$  is the number of neighbors.

Self-consistency:  $m = \tanh(\beta h_{\text{eff}}) = \tanh(\beta(h + Jzm))$ .

For  $h = 0$ :  $m = \tanh(\beta Jzm)$ . Solutions: -  $T > T_c^{MF} = Jz/k_B$ : only  $m = 0$  (disordered) -  $T < T_c^{MF}$ : nonzero  $\pm m$  solutions (ordered)

Near  $T_c$ :  $m \propto (T_c - T)^{1/2}$  — mean-field critical exponent  $\beta = 1/2$ .

Mean-field is exact for infinite-range models, qualitatively correct for high dimensions, but quantitatively wrong for low-D systems with strong fluctuations. Renormalization group (Chapter 9) systematically corrects mean-field.

#### 8.5 Landau theory: symmetry-breaking

Landau 1937: phenomenological framework for phase transitions. Free energy expansion in powers of order parameter  $\phi$ :

$$F(\phi, T) = F_0(T) + \frac{1}{2}a(T)\phi^2 + \frac{1}{4}b\phi^4 + \dots$$

with  $a(T) = a_0(T - T_c)$  changing sign at  $T_c$ .

For  $T > T_c$ :  $a > 0$ , minimum at  $\phi = 0$  (disordered). For  $T < T_c$ :  $a < 0$ , minimum at  $\phi = \pm\sqrt{-a/b}$  (ordered, symmetry broken).

Landau's framework gives the universal mean-field critical exponents and provides a phenomenological basis for understanding broad classes of transitions.

Substrate reading: Landau's  $\phi^4$  theory is the substrate's K-type cluster amplitude effective theory near  $T_c$ . The cubic-quartic balance is universal because it's the substrate's natural low-order expansion structure.

#### 8.6 First-order transitions

For first-order transitions, the order parameter jumps from one value to another at  $T_c$ . Two phases coexist along the transition line (e.g., water and steam coexist at 100°C, 1 atm).

Clausius-Clapeyron (Chapter 4 Section 4.8):  $dP/dT = L/(T\Delta V)$  along the coexistence curve.

Latent heat:  $L = T(S_2 - S_1)$  — the energy needed to convert one phase to the other at constant  $T, P$ .

Substrate reading: first-order is the substrate's K-type cluster *discontinuous* reorganization — the two phases correspond to distinct K-type configurations with finite energy gap; thermodynamic potential has two equal minima at  $T_c$ .

### 8.7 Critical point: where first-order becomes second-order

For liquid-gas: the coexistence line ends at a critical point ( $T_c, P_c$ ) where liquid and gas become indistinguishable. Beyond the critical point: continuous transition (no phase separation).

For water:  $T_c = 647$  K,  $P_c = 22.1$  MPa. For  $\text{CO}_2$ :  $T_c = 304$  K,  $P_c = 7.4$  MPa.

Near the critical point: critical phenomena dominate (Chapter 9). Density fluctuations diverge; correlation length  $\xi \rightarrow \infty$ ; substance becomes opalescent (critical opalescence).

### 8.8 Universality classes

Different physical systems can share critical exponents if they share: - Order parameter dimensionality - Spatial dimensionality - Range of interactions (short vs long) - Underlying symmetries

Universality classes: - 2D Ising: scalar order parameter,  $d = 2$  — exponents  $\beta = 1/8, \gamma = 7/4, \nu = 1$  - 3D Ising: scalar order parameter,  $d = 3$  — exponents  $\beta \approx 0.326, \gamma \approx 1.237, \nu \approx 0.630$  - 3D XY: 2-component order parameter — superfluid helium, superconductors - 3D Heisenberg: 3-component order parameter — ferromagnets

Liquid-gas critical point belongs to 3D Ising class (scalar density order parameter).

Substrate reading: universality is the substrate's K-type cluster behavior being insensitive to microscopic details — what matters is the cluster symmetry and dimensionality.

### 8.9 Worked example: 2D Ising critical exponents

For 2D Ising: - Order parameter:  $m \propto (T_c - T)^{1/8}$  as  $T \rightarrow T_c^-$  (Onsager 1944) - Susceptibility:  $\chi \propto |T - T_c|^{-7/4}$  - Specific heat:  $C \propto -\ln |T - T_c|$  (logarithmic divergence) - Correlation length:  $\xi \propto |T - T_c|^{-1}$

These exponents are exact (Onsager). Mean-field would give  $\beta = 1/2, \gamma = 1$ , etc. — completely different from the true 2D values. Mean-field fails because 2D has strong fluctuations not captured by neighbor-mean averaging.

Substrate reading: in 2D the substrate's K-type cluster fluctuations are strong and non-Gaussian; mean-field's neighbor-averaging misses them. The RG (Chapter 9) systematically incorporates fluctuations.

### 8.10 Spontaneous symmetry breaking

When the order parameter is multi-component (e.g., 3D ferromagnet  $\vec{M}$ ), the ordered phase chooses a specific direction in order-parameter space — the symmetry is *spontaneously broken*. The system's microscopic Hamiltonian has full rotational symmetry, but the ground state breaks it.

Consequences: - Goldstone bosons: massless excitations for continuous broken symmetries (spin waves in ferromagnets, phonons in crystals — though phonons are subtler) - Topological defects: vortices, domain walls, monopoles — singularities in the order parameter

Substrate reading: spontaneous symmetry breaking is the substrate's K-type cluster choosing one of multiple equivalent ground states. The Goldstone modes are the substrate's residual continuous K-type-cluster degrees of freedom.

### 8.11 K-audit anchors

- Casey's Principle: phase transitions are substrate counting transitions
- Chapter 5 of this volume: partition function (computational engine for  $F$ , hence for phase boundaries)
- Chapter 9 of this volume: critical phenomena and RG (next step)
- Volume 9 Chapter 5: topological phases (special class of phase transitions)

### Level 3 — 5th-grader accessibility

Water at 100°C boils. Iron at 770°C loses its magnetism (Curie point). Helium at 2.17 K becomes a superfluid. These are phase transitions — sudden changes in how matter is organized. Order parameters track the change: magnetization for ferromagnets, density difference for liquid-gas, condensate amplitude for superfluids. First-order transitions (like boiling water) have latent heat — energy is absorbed without temperature change. Second-order transitions (like ferromagnetic Curie point) have no latent heat but show critical phenomena (next chapter). The amazing thing: very different systems can share the same critical behavior (universality classes). Water near its critical point and a 3D Ising magnet near  $T_c$  behave identically. In BST, phase transitions are the substrate's K-type clusters reorganizing — and universality reflects the substrate's indifference to microscopic details.

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### What comes next

Chapter 9 develops critical phenomena and the renormalization group — including BST's 7-step cyclotomic cascade reading of RG (K59 ratified).

### Where to look this up

- Ising model: Ising 1925; Onsager 1944
- Landau theory: Landau 1937; Landau and Lifshitz, *Statistical Physics*
- Universality: Kadanoff 1966; Stanley, *Introduction to Phase Transitions and Critical Phenomena*
- BST anchors: Casey's Principle; Volume 9 topological phases